



MALLA REDDY TECHNICAL CAMPUS
MAISAMMAGUDA, MEDCHAL, HYDERABAD-500100
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.TECH III YEAR II SEMESTER REGULAR EXAMINATIONS

Branch / Dept: Computer Science and Engineering	Course Code: R20-CSE	Subject Code: CS601PC
Subject: Design and Analysis of Algorithms	Date of Exam: 15-11-2026	Max Marks: 25 Marks
Semester / Year: III Year II Semester	Academic Year: 2026-2027	Time / Duration: 1:30Hours

General Instructions:

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

SECTION I

SHORT ANSWERS

Instructions: Answer all 5 questions. [5 Questions × 2M | Attempt any 5 | Max Marks: 10]

1. Explain the mathematical or logical formulation underlying Introduction: Algorithm. [2 Marks]
2. How is Disjoint Sets: Disjoint set operations implemented in modern software systems? Explain with pseudocode or schema. [2 Marks]
3. Compare and contrast the advantages and limitations of Dynamic Programming: General method. [2 Marks]
4. Explain the mathematical or logical formulation underlying Greedy method: General method. [2 Marks]
5. Compare and contrast the advantages and limitations of Branch and Bound: General method. [2 Marks]

LONG ANSWERS

Instructions: Answer any 3 questions out of 5. [5 Questions × 5M | Attempt any 3 | Max Marks: 15]

6. Illustrate the step-by-step mechanism of Performance Analysis-Space complexity using a concrete example. [5 Marks]
7. Explain the fundamental working principle of union and find algorithms with suitable illustrations. [5 Marks]
8. How is applications- Optimal binary search tree implemented in modern software systems? Explain with pseudocode or schema. [5 Marks]
9. Describe the architectural or structural flow involved in applications- Job sequencing with deadlines. [5 Marks]
10. Compare and contrast the advantages and limitations of applications. [5 Marks]